

Deep Learning-Enabled Angle Estimation in Bistatic ISAC Systems

Salmane Naoumi*, Ahmad Bazzi[†], Roberto Bomfin[†] and Marwa Chafii*[†]

*NYU WIRELESS, NYU Tandon School of Engineering, New York, USA

[†]Engineering Division, New York University (NYU), Abu Dhabi, UAE.

Abstract—Integrated sensing and communication (ISAC) is becoming a vital technology for future wireless systems and is particularly relevant for many applications requiring both high-performance sensing and wireless communications. Our work presents a novel algorithm based on deep learning for estimating angles of arrival and angles of departure in bistatic ISAC systems, exploiting orthogonal frequency division multiplexing signals initially designed for communication purposes. Our proposed method incorporates a complex-valued neural network that takes advantage of the estimated channel matrix, along with a preprocessing step for coarse timing estimation. Comparative analysis with an adapted version of the multiple signal classification method demonstrates the effectiveness of our approach, showcasing remarkable performance in terms of mean squared error while requiring lower computational complexity.

Index Terms—Integrated sensing and communication (ISAC), bistatic radar, deep learning (DL), angle of arrival (AoA) estimation, angle of departure (AoD) estimation.

I. INTRODUCTION

Over the last few years, integrated sensing and communication (ISAC) has emerged a game-changing paradigm, making it possible to jointly design radar sensing and communication systems [1], [2], [3] and therefore improving the spectrum efficiency and reducing the hardware cost and computational resources. Indeed, ISAC capitalizes on the remarkable similarities in signal processing and system design between both radar and wireless communication systems and makes a compelling case for their joint design, creating new possibilities for groundbreaking applications in various fields, including automotive technology, Internet of things (IoT), and robotics. Communication-centric ISAC is an emerging field of research that shows promise in enhancing sensing capabilities through the use of optimized communication waveforms. By repurposing the estimation process of the channel state information (CSI), it becomes possible to accurately determine physical sensing parameters including the angle, range, and speed of targets of interest. Numerous research studies have investigated the use of bistatic radar systems in conjunction with orthogonal frequency-division multiplexing (OFDM)-based signal processing to optimize communication-aided sensing [4]–[6] and have validated the effectiveness of the approach within practical bistatic setups [7]. However, achieving high-quality sensing by processing the received communication signal poses practical challenges as both communication and radar systems are not mutually optimized for the sensing function. These challenges encompass handling multi-path reflections,

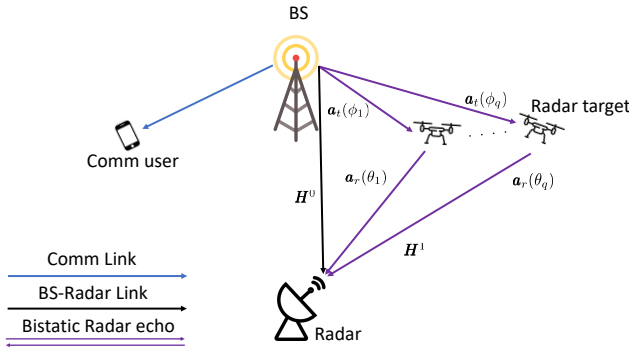
mitigating clutter noise, optimizing side-lobes and tackling Doppler ambiguity caused by the cyclic prefix (CP) [8].

In our study, we present a novel method addressing the sensing estimation problem that capitalizes on the CSI as input and trains complex-valued neural networks (NNs) to minimize the mean squared error (MSE) objective function on training datasets generated using the ideal system model. Moreover, we benchmark our algorithm with a 2D-multiple signal classification (MUSIC) algorithm that exploits the knowledge of the model. More specifically, the MUSIC method simultaneously estimates the angle of arrival (AoA) and angle of departure (AoD) of the targets exploiting full knowledge of the transmit/receive arrays, which is further used to test for noise subspace orthogonality, as per the MUSIC principle. The results reveal that the NN-based solution provides a good estimation performance while requiring considerably less computational effort. The contributions of this work are summarized as follows.

- Development of an NN algorithm to estimate the AoA and AoD based on the estimated channel matrix.
- Design of a benchmark algorithm adapting the MUSIC algorithm for the bistatic ISAC system model.
- Complexity analysis of both methods in terms of the number of multiplications and additions.
- Performance comparison in terms of MSE and complexity between the NN-based estimation method and the MUSIC-based method.

The detailed structure of the following paper is given as follows. In Section II, we introduce the communication-centric ISAC system model, along with the channel estimator and coarse timing estimate used throughout the paper. In Section III, we present our complex-valued NN-based AoA and AoD estimator. Additionally, we present the adapted 2D-MUSIC algorithm in Section IV. Then, we provide a computational complexity analysis of both algorithms in Section V. Finally, Section VI provides a performance comparison of both algorithms before concluding the paper in Section VII.

Notation: Upper-case and lower-case boldface letters denote matrices and vectors, resp. $(\cdot)^T$, $(\cdot)^*$ and $(\cdot)^H$ represent the transpose, the conjugate and the transpose-conjugate operators. For any complex number $z \in \mathbb{C}$, the real part of z is denoted as $\Re(z)$, whereas the imaginary part is denoted as $\Im(z)$. The ℓ_2 norm of a vector $\mathbf{x}$ is denoted as $\|\mathbf{x}\|$. The matrices $\mathbf{F}$ and $\mathbf{I}$ are the Fourier and the identity matrices with appropriate


 Fig. 1. An ISAC scenario composed of q targets in a bistatic fashion.

dimensions, resp. For matrix indexing, the $(i, j)^{th}$ entry of matrix $\mathbf{A}$ is denoted by $[\mathbf{A}]_{[i,j]}$ and its j^{th} column is denoted as $\mathbf{A}_{[:,j]}$. The operator $\otimes$ is the *Kronecker* product. The big- $\mathcal{O}$ notation is $\mathcal{O}()$.

II. SYSTEM MODEL

Consider a base station (BS) with N_t antennas transmitting OFDM symbols in the downlink. The OFDM symbol duration, excluding the CP, is denoted as $T = \frac{1}{\Delta_f}$, where Δ_f is the sub-carrier spacing. Before expressing the transmit symbols, we describe the wireless channel related to the bistatic radar. The system model is represented in Fig. 1.

A. Bistatic Radar Channel Model

Under the simplifying assumptions of no clutter, no frequency and time offsets between the BS and the radar unit in addition to non-moving targets, the general channel impulse response (CIR) between the n_t^{th} transmit antenna at the BS and the n_r^{th} receive antenna at the radar unit is expressed as

$$h_{n_r, n_t}(t, \tau) = \sum_{m=0}^q \alpha_m a_{n_r}(\theta_m) a_{n_t}(\phi_m) \delta(\tau - \tau_m), \quad (1)$$

where $n_r = 1 \dots N_r$ and $n_t = 1 \dots N_t$ for N_r receive antennas, and N_t transmit antennas. In addition, α_m and τ_m denote the signal attenuation coefficient and delay of the m^{th} reflection, respectively, and are assumed to be time-invariant during the parameter estimation period. The function $\delta(\cdot)$ is the Dirac delta function, and q is the number of targets. Furthermore, θ_m represents the AoA between the m^{th} target and the radar unit, and ϕ_m is the AoD between the BS and the m^{th} target. Also, $a_{n_r}(\theta)$ is the steering coefficient at the n_r^{th} receive antenna at the radar due to AoA θ and $a_{n_t}(\phi)$ is the steering coefficient at the n_t^{th} transmit antenna at the BS due to AoD ϕ . Note that the first path, i.e. $m = 0$, represents the line of sight (LoS) path between the BS and the radar unit. The channel given in (1) can be expressed in the discrete frequency domain on the n^{th} sub-carrier, in matrix form, as

$$\mathbf{H}_n = \mathbf{A}_r(\Theta) \mathbf{G} \mathbf{D}_n(\tau) \mathbf{A}_t^T(\Phi) \in \mathbb{C}^{N_r \times N_t}. \quad (2)$$

In (2), the matrix $\mathbf{A}_r(\Theta) = [\mathbf{a}_r(\theta_0) \ \mathbf{a}_r(\theta_1) \ \dots \ \mathbf{a}_r(\theta_q)]$ is the steering matrix resulting from angles of arrival (AoAs)

and $\mathbf{a}_r(\theta)$ is the steering vector of the radar's antenna elements. Likewise, $\mathbf{A}_t(\Phi) = [\mathbf{a}_t(\phi_0) \ \mathbf{a}_t(\phi_1) \ \dots \ \mathbf{a}_t(\phi_q)]$ represents the steering matrix due to angles of departure (AoDs), where $\mathbf{a}_t(\phi)$ is the steering vector arising from the BS antenna elements. Moreover, $\mathbf{G} = \text{diag}([\alpha_0 \ \dots \ \alpha_q])$ contains the signal attenuation coefficients. Also, $\mathbf{D}_n(\tau) = \text{diag}([c_n(\tau_0) \ \dots \ c_n(\tau_q)])$ is a matrix containing all delays $c_n(\tau) = e^{-j2\pi n \Delta_f \tau}$, which represents the delay response at the n^{th} sub-carrier. In the next sub-section, we describe the received signal in the frequency domain as seen by the radar.

B. Radar Signal Model

The BS transmits K OFDM signals serving communication users in the scene. The k^{th} OFDM can be expressed as

$$\mathbf{x}_k(t) = \sum_{n=1}^N \mathbf{s}_{n,k} c_n(-t) \Pi(t - kT_o), \quad \forall k = 1 \dots K, \quad (3)$$

where N is the number of active sub-carriers occupying each of the K OFDM symbols and $T_o = T + T_{CP}$ is the overall OFDM symbol duration. Moreover, $\Pi(t)$ is the windowing function. We assume an ideal rectangular function, that is

$$\Pi(t) = \begin{cases} 1, & t \in [-T_{CP}, T] \\ 0, & \text{otherwise,} \end{cases} \quad (4)$$

where T_{CP} is the CP duration, which should be greater than the maximum of propagation delays in order to guarantee a cyclic convolution with the channel. Furthermore, $\mathbf{s}_{n,k} \in \mathbb{C}^{N_r \times 1}$ is the modulated symbol onto the n^{th} OFDM sub-carrier within the k^{th} OFDM symbol. Combining (2), (3), (4) and applying fast Fourier transform (FFT), the radar unit reads the following data on the n^{th} subcarrier and k^{th} OFDM symbol,

$$\mathbf{y}_{n,k} = \mathbf{H}_n \mathbf{s}_{n,k} + \mathbf{w}_{n,k} \in \mathbb{C}^{N_r \times 1}. \quad (5)$$

In this paper, we focus on a communication-centric ISAC problem, where an existing infrastructure is performing communication tasks, whereas a radar unit is installed to estimate the sensing parameters of the different targets.

C. Channel Estimation

In this sub-section, we describe the channel estimates that are used for both deep learning (DL)-based and parameterized methods to estimate AoA and AoD. Assuming K_P OFDM symbols occupying N_P sub-carriers each are being transmitted, and assuming that the radar unit has knowledge about the data symbols being transmitted, the channel estimation is performed according to the well-known least squares as

$$\bar{\mathbf{H}}_n = \mathbf{Y}_n \mathbf{S}_{P,n}^H (\mathbf{S}_{P,n} \mathbf{S}_{P,n}^H)^{-1}, \quad (6)$$

where $\mathbf{Y}_n$ contains all pilot OFDM symbols, i.e. $\{\mathbf{y}_{n,k}\}_{k=1}^{K_P}$. Likewise, $\mathbf{S}_{P,n}$ contains the known pilot information, $\{\mathbf{s}_{n,k}\}_{k=1}^{K_P}$. Note that $\bar{\mathbf{H}}_n = \mathbf{H}_n + \mathbf{W}_{P,n} \mathbf{S}_{P,n}^H (\mathbf{S}_{P,n} \mathbf{S}_{P,n}^H)^{-1}$. It is more convenient to express CSI estimates as follows

$$\bar{\mathbf{H}} = [\text{vec}(\bar{\mathbf{H}}_1) \ \text{vec}(\bar{\mathbf{H}}_2) \ \dots \ \text{vec}(\bar{\mathbf{H}}_{N_P})], \quad (7)$$

where $\bar{\mathbf{H}} \in \mathbb{C}^{N_t N_r \times N_P}$ contains the frequency domain channel in its rows for all combinations of transmit/receive antenna pairs.

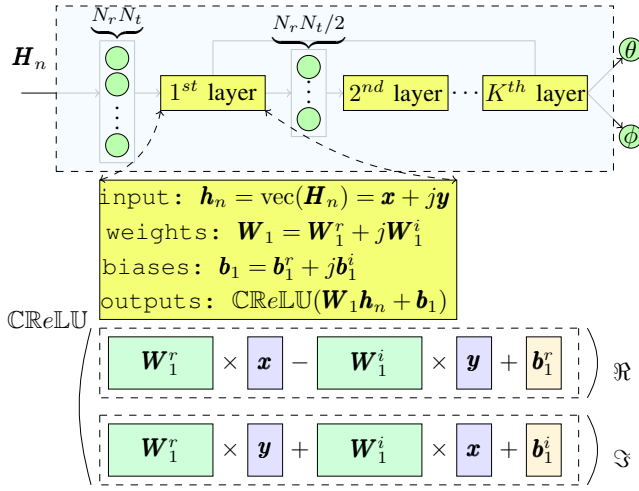


Fig. 2. Architecture of the proposed NN composed of complex-valued linear layers and CReLU activation functions. The channel matrix at each peak is used to compute the corresponding AoA and AoD.

D. Coarse Timing Estimation

The size of the channel estimation matrix in (7) can be reduced in order to estimate AoA and AoD, such that the input size of the machine learning (ML) algorithm is decreased. Since each row of $\bar{\mathbf{H}}$ in (7) represents the frequency domain channel over a given transmit and receive antenna pair, we can take the IFFT of (7) over its rows, and find the peaks corresponding to the time delay of a target reflection, which can be interpreted as a coarse timing estimation for the delay as a multiple of the sampling time. Notice that the AoA and AoD have a direct relation with the phase of the peaks in the time domain, thus, a matrix of size $N_r N_t$ with the peaks can be used to estimate these parameters instead of the larger matrix of (7). This method is briefly shown in the following. The inverse fast Fourier transform (IFFT)s per transmit/receive antenna pairs is performed as

$$\bar{\mathbf{h}}_n = \text{IFFT}(\bar{\mathbf{H}}_{[n,:]}^T) = \mathbf{F}^H \bar{\mathbf{H}}_{[n,:]}^T, \quad (8)$$

$\forall n = 1 \dots N_r N_t$. Then, the task is basically to find the indexes k corresponding to the q most likely peaks over all antenna pairs $[\bar{\mathbf{h}}_{n[k]}]$ from $\bar{\mathbf{h}}_n = [\bar{\mathbf{h}}_{n[1]} \dots \bar{\mathbf{h}}_{n[N_P]}] \in \mathbb{C}^{N_P}$, which are defined as

$$k \in \{\hat{i}_1, \dots, \hat{i}_q\}, \quad (9)$$

and are found by analyzing the peaks of all $N_r N_t$ antenna pairs. Note that a coarse estimate of the times of arrival (ToAs) is obtained by scaling the indices as $\hat{\tau}_k = \frac{1}{N_P \Delta_f} \hat{i}_k, \forall k$.

In our work, we are considering the well-separated case, i.e. when all i_k s are distinct ($i_1 \neq i_2 \neq \dots \neq i_q$). It is easily verified that the i_k -th row of $\mathbf{F}^H \bar{\mathbf{H}}^T$ is the only row that contains information about θ_k and ϕ_k . Other special cases can occur where some or all the paths, say $p \leq q$, arrive within the same bin. In that case, the corresponding i^{th} row of $\mathbf{F}^H \bar{\mathbf{H}}^T$ contains all the AoA and AoD information for the p paths and can be reshaped as

$$\hat{\mathbf{H}} = \mathbf{A}_r(\Theta) \mathbf{G} \mathbf{A}_t^T(\Phi) + \hat{\mathbf{W}}. \quad (10)$$

Note that for the well-separated case, the i_k -th row of $\mathbf{F}^H \bar{\mathbf{H}}^T$ would be $\alpha_k \mathbf{a}_r(\theta_k) \mathbf{a}_t^T(\phi_k) + \hat{\mathbf{W}}_k$, i.e. a special case of (10).

III. MACHINE LEARNING-BASED AOA AND AOD ESTIMATION

While most of the reviewed literature [9]–[11] uses the receive matrix¹, the known pilot information $\mathbf{S}_{P,n}$ or their covariance matrix as input, we leverage the estimated channel matrix to extract, with an IFFT processing step, matrices per target k , denoted by $\hat{\mathbf{H}}_k$, of reduced size $N_r N_t$ and containing all the information about the targets parameters θ_k and ϕ_k , as described in Section II-D.

In our work, we leverage complex-valued NNs [12], inspired by recent findings indicating that complex numbers possess a richer representational capacity and attractive properties. This choice is particularly relevant due to the inherent complex-value operations involved in modeling communication systems. Hence, we use complex-valued linear layers as fundamental building blocks, which are constructed using real-values linear ones due to the distributive property of the multiplication operator. Indeed, for a complex-valued NN where the weights and bias parameters of the k -th layer are $\mathbf{W}_k = \mathbf{W}_k^r + j\mathbf{W}_k^i$ and $\mathbf{b}_k = \mathbf{b}_k^r + j\mathbf{b}_k^i$, respectively. The response of this layer to an input formed from the previous layer as $\mathbf{z}^{k-1} = \mathbf{x}^{k-1} + j\mathbf{y}^{k-1}$ is given by

$$\mathbf{z}^k = (\mathbf{W}_k^r \mathbf{x}^{k-1} - \mathbf{W}_k^i \mathbf{y}^{k-1} + \mathbf{b}_k^r) + j(\mathbf{W}_k^r \mathbf{y}^{k-1} + \mathbf{W}_k^i \mathbf{x}^{k-1} + \mathbf{b}_k^i). \quad (11)$$

Moreover, we employ the complex rectified linear unit CReLU as an activation function that operates independently on the real and imaginary components of each complex neuron, i.e.

$$\text{CReLU}(\mathbf{z}^k) = \text{ReLU}(\Re(\mathbf{z}^k)) + j \text{ReLU}(\Im(\mathbf{z}^k)). \quad (12)$$

As specified in Section II, we consider the case with well-separated targets. We illustrate in Fig. 2 our architecture consisting of $K = 3$ hidden layers and a final output layer comprising of $N_{out} = 2$ neurons corresponding to the predictions for both AoA and AoD. For each target k , the input is the i_k -th peak row of the matrix $\mathbf{F}^H \bar{\mathbf{H}}^T$ of size $S_{inp} = N_r N_t$ containing information about the specific target AoA (θ_k) and AoD (ϕ_k). Then, the three hidden layers are mapping the input to the following latent dimensions $[\lfloor \frac{S_{inp}}{2} \rfloor, \lfloor \frac{S_{inp}}{4} \rfloor, \lfloor \frac{S_{inp}}{8} \rfloor]$, and lastly, we use the final layer to estimate both $\hat{\theta}_k^{\text{tar}}$ and $\hat{\phi}_k^{\text{tar}}$. For training purposes, we collect batches from E simulation experiments with q well-separated targets and we treat the estimation problem as a regression task by minimizing the MSE objective function given as

$$\text{MSE} = \frac{1}{qE} \sum_{e=1}^E \sum_{k=1}^q \left[(\hat{\theta}_k^{\text{tar}}(e) - \theta_k(e))^2 + (\hat{\phi}_k^{\text{tar}}(e) - \phi_k(e))^2 \right]. \quad (13)$$

The training process employs the backpropagation algorithm along with stochastic gradient descent and Adam optimizer and consists of 300 epochs, during which we assess the

¹In our case it corresponds to $\mathbf{Y}_n$ containing all OFDM symbols.

model's performance and investigate overfitting by analyzing the training and validation MSE losses. To address overfitting concerns, we adjust the learning rate schedule, starting at $1e^{-4}$ and decreasing it by a factor of 2 at epochs 200 and 250. It is worth noting that different values of signal-to-noise ratio (SNR) ranging from -5 to 30 dB were used to collect simulation data and train the NN so as to enhance the network's robustness, resulting in improved performance across various SNR levels and channel realizations.

IV. BENCHMARK METHOD: 2D MUSIC ALGORITHM

In this section, we present a MUSIC-inspired algorithm [13] adapted to the system model of Section II, which will be used as a benchmark for the ML-based approach presented in Section III.

Similar to the concept of MUSIC, our benchmark method exploits the eigen decomposition of the covariance matrix of the CSI $\bar{\mathbf{H}}$ described in (7). We proceed by extracting the signal and noise subspaces, taking advantage of their orthogonality to separate them. Subsequently, the joint AoA/AoD pairs are estimated by utilizing the orthogonality between the noise space and the steering vectors evaluated at those angles. More specifically, the covariance matrix of the CSI is computed as

$$\bar{\mathbf{R}} = \frac{1}{N_p} \bar{\mathbf{H}} \bar{\mathbf{H}}^H = \frac{1}{N_p} \sum_{n=1}^{N_p} \text{vec}(\bar{\mathbf{H}}_n) \text{vec}^H(\bar{\mathbf{H}}_n). \quad (14)$$

We denote by $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_{N_r N_t}$ the eigenvalues of $\bar{\mathbf{R}}$ and their corresponding orthogonal eigenvectors $\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_{N_r N_t}$. Subsequently, we construct the noise subspace matrix by selecting the eigenvectors associated with the smallest $N_r N_t - q$ eigenvalues, which is defined as

$$\mathbf{U}_n = [\mathbf{u}_{q+1} \quad \mathbf{u}_{q+2} \quad \dots \quad \mathbf{u}_{N_r N_t}]. \quad (15)$$

Then, the 2D-MUSIC spatial spectrum is computed as

$$\begin{aligned} P(\theta, \phi) &= \frac{1}{[\mathbf{a}_t(\phi) \otimes \mathbf{a}_r(\theta)]^H \mathbf{U}_n \mathbf{U}_n^H [\mathbf{a}_t(\phi) \otimes \mathbf{a}_r(\theta)]} \\ &= \frac{1}{\|\mathbf{U}_n^H [\mathbf{a}_t(\phi) \otimes \mathbf{a}_r(\theta)]\|^2}, \end{aligned} \quad (16)$$

and is evaluated at each combination of AoA and AoD (θ_i, ϕ_i) , where the resolution of the search is dependant on the grid sizes of AoA and AoD denoted by G_θ and G_ϕ . Finally, the angles are obtained by searching for the peaks of the 2D spatial spectrum. The implementation steps of the MUSIC algorithm are summarized in **Algorithm 1**.

Algorithm 1 2D-MUSIC Algorithm for bistatic ISAC

INPUT: $\mathbf{Y}_P, \mathbf{S}_P$

CHANNEL ESTIMATION:

Obtain $\{\hat{\mathbf{H}}_n\}_{n=1}^{N_p}$ according to equation (6).

Construct the CSI $\bar{\mathbf{H}}$ according to equation (7).

SENSING ESTIMATION:

1) Obtain the covariance matrix $\bar{\mathbf{R}}$ of the CSI according to equation (14).

2) Get the eigenvalues of $\bar{\mathbf{R}}$, i.e. $\{\lambda_i\}_{i=1}^{N_r N_t}$ and corresponding eigenvectors $\{\mathbf{u}_i\}_{i=1}^{N_r N_t}$.

3) Form the noise subspace matrix $\mathbf{U}_n$ as described in equation (15).

4) For each combination of (θ_i, ϕ_i) defined by the grid sizes G_θ and G_ϕ , evaluate the 2D spatial spectrum $P(\theta_i, \phi_i)$ as advised by equation (16).

5) Extract the sensing parameters by analyzing the q peaks of $\{P(\theta, \phi)\}_{(\theta, \phi) \in \mathcal{G}}$ over the entire grid space $\mathcal{G}$.

return $(\hat{\theta}_1, \hat{\phi}_1) \dots (\hat{\theta}_q, \hat{\phi}_q)$.

V. COMPUTATIONAL COMPLEXITY ANALYSIS

In this section, we provide an analysis of the computational complexity of the NN and MUSIC algorithms. Notably, we exclude the complexity of the channel estimation block discussed in II-C, as it is shared by both algorithms and thus excluded from the analysis. The computational complexity of both algorithms is derived using the following rules

- 1) AB where $A \in \mathbb{C}^{m \times n}$ and $B \in \mathbb{C}^{n \times l}$ requires mn multiplications and $m(n-1)l$ additions.
- 2) Eigendecomposition of $A \in \mathbb{C}^{n \times n}$ requires $9n^2(n-1)$ multiplications, $9n^2(n-1)$ additions and $2n(n-1)$ square roots assuming both eigenvectors and eigenvalues are of interest.
- 3) Using a coordinate rotation digital computer (CORDIC) [14] with a number of iterations N_{cord} and ignoring the shift operations
 - A phase retrieval operation requires $2N_{\text{cord}}$ additions and 2 multiplications.
 - A square root operation costs $2N_{\text{cord}}$ additions and 1 multiplication.

A. Complex-valued NN

To accommodate the input for the NN algorithm, an additional preprocessing step is done to extract peaks from the IFFT of the estimated channel matrix. This process involves retrieving peaks across all antenna pairs, as outlined in (8). This requires matrix multiplication of $\mathbf{F}^H \in \mathbb{C}^{N_r N_t \times N_r N_t}$ with the transpose of the estimated channel matrix $\bar{\mathbf{H}}^T \in \mathbb{C}^{N_r N_t \times N_p}$, which can be done with an FFT algorithm costing $N_p N_r N_t \log(N_r N_t)$ multiplications and $N_p N_r N_t \log(N_r N_t)$ additions. Note that the peak extraction task is not considered since it does not require any floating-point operations. At inference time, the computational complexity of the NN in terms of the total number of additions and multiplications is obtained by summing up the contributions of all the layers.

Assuming each layer is mapping from $\mathcal{S}_i$ to $\mathcal{S}_o$ with a CReLU , then a total of $4\mathcal{S}_o\mathcal{S}_i + 2\mathcal{S}_o$ multiplications and $4\mathcal{S}_o(\mathcal{S}_i - 1) + 2\mathcal{S}_o$ additions is required as per the equations (11) and (12) and assuming that CReLU activation function requires at most one multiplication. It is worth noting that the output layer of the network does not implement CReLU but rather a phase retrieval operation and a multiplication by $180/\pi$ to retrieve both AoA and AoD, which costs a total of $2N_{\text{cord}}$ additions and 2 multiplications. Therefore the total computational cost of the NN, assuming the input size $\mathcal{S}_{\text{inp}} = N_r N_t$ and $\lfloor \frac{\mathcal{S}_{\text{inp}}}{2} \rfloor, \lfloor \frac{\mathcal{S}_{\text{inp}}}{4} \rfloor, \lfloor \frac{\mathcal{S}_{\text{inp}}}{8} \rfloor$ as sizes of the three hidden layers, is computed as follows

$$\begin{aligned} T_{\text{mul}} &= 4 \left(\mathcal{S}_{\text{inp}} \lfloor \frac{\mathcal{S}_{\text{inp}}}{2} \rfloor + \lfloor \frac{\mathcal{S}_{\text{inp}}}{2} \rfloor \lfloor \frac{\mathcal{S}_{\text{inp}}}{4} \rfloor + \lfloor \frac{\mathcal{S}_{\text{inp}}}{4} \rfloor \lfloor \frac{\mathcal{S}_{\text{inp}}}{8} \rfloor \right) \\ &\quad + 2 \left(\lfloor \frac{\mathcal{S}_{\text{inp}}}{2} \rfloor + \lfloor \frac{\mathcal{S}_{\text{inp}}}{4} \rfloor + 5 \lfloor \frac{\mathcal{S}_{\text{inp}}}{8} \rfloor + 1 \right) \\ &\quad + N_p \mathcal{S}_{\text{inp}} \log(\mathcal{S}_{\text{inp}}) \\ T_{\text{add}} &= 4 \left(\mathcal{S}_{\text{inp}} \lfloor \frac{\mathcal{S}_{\text{inp}}}{2} \rfloor + \lfloor \frac{\mathcal{S}_{\text{inp}}}{2} \rfloor \lfloor \frac{\mathcal{S}_{\text{inp}}}{4} \rfloor + \lfloor \frac{\mathcal{S}_{\text{inp}}}{4} \rfloor \lfloor \frac{\mathcal{S}_{\text{inp}}}{8} \rfloor \right) \\ &\quad - 2 \left(2\mathcal{S}_{\text{inp}} + \lfloor \frac{\mathcal{S}_{\text{inp}}}{2} \rfloor + \lfloor \frac{\mathcal{S}_{\text{inp}}}{4} \rfloor + 5 \lfloor \frac{\mathcal{S}_{\text{inp}}}{8} \rfloor - N_{\text{cord}} - 1 \right) \\ &\quad + N_p \mathcal{S}_{\text{inp}} \log(\mathcal{S}_{\text{inp}}), \end{aligned}$$

where T_{add} and T_{mul} are the total number of additional and multiplications, respectively. Note that in the case of an environment with q well-resolved targets, the computational complexity is multiplied by a factor of q due to the increased batch size of the input, which corresponds to the number of targets present in the scene.

B. MUSIC algorithm

The computational complexity of the MUSIC algorithm can be detailed per block as follows

- The computation of $\bar{\mathbf{R}}$ involving $\bar{\mathbf{H}} \in \mathbb{C}^{N_r N_t \times N_p}$ via equation (14) requires $(N_r N_t)^2 N_p$ multiplications and $(N_r N_t)^2 (N_p - 1)$ additions.
- The eigendecomposition of the covariance matrix $\bar{\mathbf{R}} \in \mathbb{C}^{N_r N_t \times N_r N_t}$ requires $9(N_r N_t)^3 - 7(N_r N_t)^2 - 2N_r N_t$ multiplications and $9(N_r N_t)^3 + (4N_{\text{cord}} - 9)(N_r N_t)^2 - 4N_{\text{cord}} N_r N_t$ additions.
- Computing $\mathbf{U}_n \mathbf{U}_n^H$ prior to evaluating the spatial spectrum requires $(N_r N_t)^2 (N_r N_t - q)$ multiplications and $(N_r N_t)^2 (N_r N_t - q - 1)$ additions.
- Evaluation of the 2D spatial spectrum over the entire 2D grid space $\mathcal{G}$ defined by grid sizes G_θ and G_ϕ for AoA and AoD requires a matrix multiplication as advised in (16) for every element of the grid costing a total of $G_\theta G_\phi ((N_r N_t)^2 + N_r N_t)$ multiplications and $G_\theta G_\phi ((N_r N_t)^2 - 1)$ additions.

To sum up, we have the following total number of operations for the MUSIC algorithm with $\mathcal{S}_{\text{inp}} = N_r N_t$

$$\begin{aligned} T_{\text{mul}} &= 10\mathcal{S}_{\text{inp}}^3 + (N_p + G_\theta G_\phi - q - 7) \mathcal{S}_{\text{inp}}^2 \\ &\quad + (G_\theta G_\phi - 2) \mathcal{S}_{\text{inp}} \end{aligned}$$

$$\begin{aligned} T_{\text{add}} &= 10\mathcal{S}_{\text{inp}}^3 + (N_p + 4N_{\text{cord}} + G_\theta G_\phi - q - 11) \mathcal{S}_{\text{inp}}^2 \\ &\quad - 4N_{\text{cord}} \mathcal{S}_{\text{inp}} - G_\theta G_\phi, \end{aligned}$$

where T_{mul} and T_{add} are the total number of multiplications and additions respectively, excluding the complexity involved in finding the peaks of the 2D spatial spectrum.

VI. NUMERICAL EVALUATION

In this work, we consider $N_t = 8$ antennas and $N_r = 10$ antennas arranged in a uniform linear array (ULA) configuration with a half-wavelength spacing $\frac{\lambda}{2}$. A bandwidth of 61.44 MHz is used. The number of active subcarriers per symbol is $N_P = 64$ and the number of OFDM symbols transmitted by the BS is $K_P = 10$, assumed to be known at the radar unit.

In Fig. 3, we provide the findings of our complexity study, which shows the overall number of operations required as a function of the product $N_r N_t$ assuming a number of targets $q = 3$ and $N_{\text{cord}} = 10$. With an increase in the system configuration, characterized by a larger product of $N_r N_t$, we observe a slower rate of increase in the total number of operations needed for the NN algorithm compared to MUSIC. It is noteworthy that this difference in computational complexity becomes particularly noticeable when smaller grid sizes are considered for the spatial spectrum search. Indeed, for a reasonable grid resolution and considering that both grids sizes are of the same order G , then the complexity of MUSIC $S = \mathcal{O}(\mathcal{S}_{\text{inp}}^2 G^2)$ is dominated by the size of G , which will be for example in the order of 10^{10} for a grid size of 0.1. The difference in the rate of increase emphasizes the advantages of employing an NN approach in terms of computational efficiency and scalability, rendering it particularly suitable for a rate with limited resources.

In terms of training computational cost, the implemented NN consists of 8584 trainable parameters, including weights and biases across different layers. For a forward pass with a batch size of 128, the number of multiply-accumulate operations (MACs) is approximately 4.321 MMACs. Considering that training is done using 50,000 data samples, the total number of MACs amounts roughly to 1.562 GMACs. Additionally, backpropagation, which consists of updating the weights of the NN, requires approximately three times more MACs than forward passes. Consequently, the total cost for training 300 epochs sums up to 1.875 TMACs. It is worth noting that training time and number of floating-point operations directly correlate with the required MACs, assuming optimal parallel execution and no slowdowns caused by data-loading processes.

We report in Fig. 4 the performance behavior of the proposed NN algorithm in comparison to MUSIC in terms of MSE to evaluate its accuracy for sensing estimation. The comparison is done over channel realizations with SNR values ranging from -9 to 30 dB as well as random AoAs and AoDs, which fall within the range $[20^\circ, 40^\circ]$ with up to 4 decimal precision. In fact, different grid sizes G_θ and G_ϕ are considered for the MUSIC algorithm, dictating the minimum

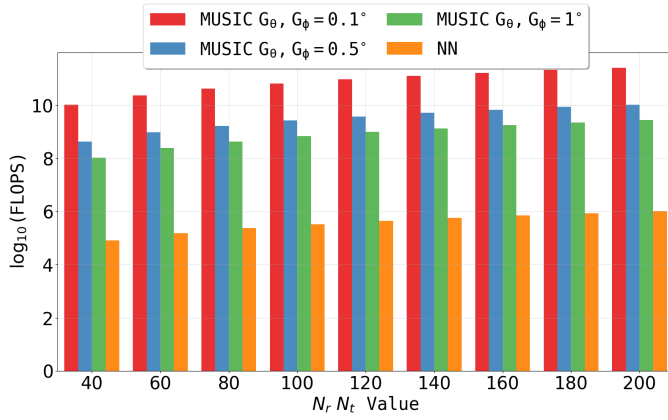


Fig. 3. Complexity comparison in terms of the total number of multiplications and additions for both NN and MUSIC estimation algorithms.

resolution that can be achieved. This explains the performance of the MUSIC algorithm remaining constant after 0 dB SNR due to discretization errors. Overall, the results obtained indicate a good performance of the NN algorithm compared to MUSIC for both AoA and AoD under various SNR regimes. For example, the NN algorithm requires an SNR greater or equal to 9 dB for a requirement of 10^{-6} MSE per radian². However, it is worth mentioning that when the SNR is high, the NN algorithm does not offer significant performance benefits, as the MSE does not decrease correspondingly. As a result, additional research and optimization are required to improve performance in such scenarios. Additionally, sub-optimality can arise from timing estimation, as the coarse estimates can affect performance at the expense of reduced complexity. Furthermore, it is important to mention that while MUSIC relies on prior knowledge of the number of targets in the scene, the NN exhibits good performance without this prerequisite. Instead, the NN method assumes that targets are well-resolved in the time domain. However, this assumption can become problematic when the temporal resolution is inadequate. In such cases, peaks from the IFFT of the estimated channel matrix can be spatially resolved into multiple targets.

VII. CONCLUSION

In this paper, a complex-valued NN-based algorithm is proposed to estimate both AoA and AoD for bistatic ISAC systems. The proposed method exploits the estimated channel matrix and incorporates a preprocessing step involving coarse timing estimation, resulting in a reduced input size and improved computational efficiency. A benchmarking analysis with a MUSIC algorithm adapted to the system model highlights the potential practical application of our algorithm for sensing estimation as it showcases the competitive performance and lower computational complexity in terms of the required number of additions and multiplications. Future work directions may include the estimation of other sensing parameters such as time of arrival and Doppler jointly with directions of departure and arrival in more realistic system

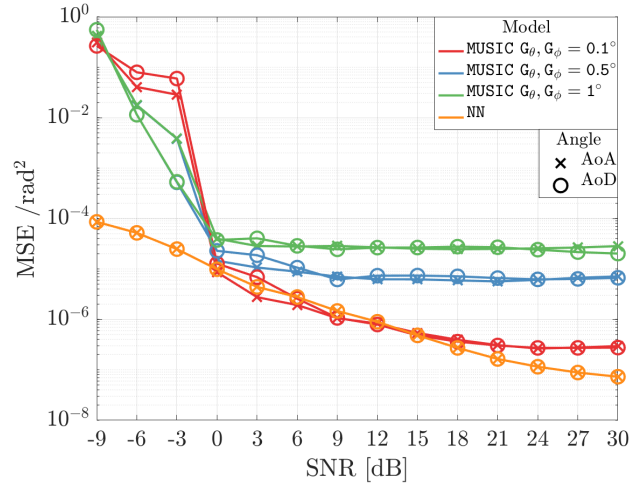


Fig. 4. Performance analysis in terms of the MSE of NN model in comparison to the MUSIC with different grid sizes [1° , 0.5° , 0.1°].

models. Achieving this would necessitate the design of more advanced NN models.

REFERENCES

- [1] F. Liu *et al.*, "Integrated sensing and communications: Toward dual-functional wireless networks for 6g and beyond," *IEEE J. Sel. Areas Commun.*, vol. 40, no. 6, pp. 1728–1767, 2022.
- [2] A. Bazzi and M. Chafii, "On Outage-Based Beamforming Design for Dual-Functional Radar-Communication 6G Systems," *IEEE Trans. Wireless Commun.*, vol. 22, no. 8, pp. 5598–5612, 2023.
- [3] A. Bazzi and M. Chafii, "On Integrated Sensing and Communication Waveforms with Tunable PAPR," *IEEE Trans. Wireless Commun.*, pp. 1–1, 2023.
- [4] R. S. Thoma *et al.*, "Cooperative Passive Coherent Location: A Promising 5G Service to Support Road Safety," *IEEE Commun. Mag.*, vol. 57, no. 9, pp. 86–92, 2019.
- [5] A. Bazzi and M. Chafii, "RIS-Enabled Passive Radar towards Target Localization," *arXiv preprint arXiv:2210.11887*, 2022.
- [6] L. Leyva, D. Castanheira, A. Silva, and A. Gameiro, "Two-stage estimation algorithm based on interleaved OFDM for a cooperative bistatic ISAC scenario," in *2022 IEEE 95th Vehicular Technology Conference: (VTC2022-Spring)*, pp. 1–6, 2022.
- [7] S. Schieler *et al.*, "OFDM waveform for distributed radar sensing in automotive scenarios," *International Journal of Microwave and Wireless Technologies*, vol. 12, pp. 1–7, 07 2020.
- [8] C. Sturm and W. Wiesbeck, "Waveform Design and Signal Processing Aspects for Fusion of Wireless Communications and Radar Sensing," *Proc. IEEE*, vol. 99, no. 7, pp. 1236–1259, 2011.
- [9] J. Fuchs *et al.*, "A Machine Learning Perspective on Automotive Radar Direction of Arrival Estimation," *IEEE Access*, vol. PP, pp. 1–1, 01 2022.
- [10] J. Fuchs, R. Weigel, and M. Gardill, "Single-snapshot direction-of-arrival estimation of multiple targets using a multi-layer perceptron," in *2019 IEEE MTT-S International Conference on Microwaves for Intelligent Mobility (ICMIM)*, pp. 1–4, 2019.
- [11] B. Milovanovic, M. Agatonovic, Z. Stankovic, N. Doncov, and M. Sarevska, "Application of neural networks in spatial signal processing (invited paper)," in *11th Symposium on Neural Network Applications in Electrical Engineering*, pp. 5–14, 2012.
- [12] C. Trabelsi *et al.*, "Deep complex networks," *arXiv preprint arXiv:1705.09792*, 2017.
- [13] R. Schmidt, "Multiple emitter location and signal parameter estimation," *IEEE Transactions on Antennas and Propagation*, vol. 34, no. 3, pp. 276–280, 1986.
- [14] S. Wang and E. Swartzlander, "Merged CORDIC algorithm," in *1995 IEEE International Symposium on Circuits and Systems (ISCAS)*, vol. 3, pp. 1988–1991 vol.3, 1995.